

Project Demo Planning

Date	
Team ID	
Project Name	OptiCrop - Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Project Demo Planning:

A well-structured demo plan ensures that the team presents the project effectively, covering all key aspects in a clear and organized manner. This document outlines the plan for demonstrating the project, including the flow of the demo, key features to highlight, and responsibilities of each team member.

S.No	Demo Section	Description	Duration (mins)	Responsible Member
1	Introduction & Problem Statement - Overview of agricultural challenges - Need for smart crop recommendation - Project objectives and scope	3 min	Ramya Sri Dasari	1
2	Dataset & Exploratory Data Analysis - Dataset overview (2200 records, 22 crops) - Univariate/Bivariate/Multivariate analysis - Key insights and visualizations	3 min	Sai Naga Hansika Mukkollu	2
3	Data Pre-processing & Model Building - Handling null values and outliers - Train/test split & feature engineering - 6 ML models trained & compared - K-Means clustering with PCA	4 min	Anumakonda Janakiram	3
4	Live Feature Demonstration - Crop Recommendation (form input to result)	5 min	Amarthala Kanthi Sri	4

	- Suitability Analysis (full assessment) - Research Dashboard (charts, stats, model comparison)			
5	Architecture & Deployment - Frontend (Vercel) + Backend (Render) - REST API endpoints overview - Live deployed application walkthrough	3 min	Baby Jahnavi Madiseti	5
6	Conclusion & Q&A - Summary of achievements - Challenges faced and solutions - Future enhancements - Open floor for questions	2 min	All team members	6

Demo Flow Summary:

Step	Activity	Notes
1	Introduction & Problem Statement (Ramya Sri Dasari)	Explain the agricultural productivity challenge. State that 60% of Indian farmland lacks scientific crop planning. Present OptiCrop as an ML-powered solution.
2	Solution Overview (Baby Jahnavi Madiseti)	Show project architecture: frontend (Vercel) communicates with backend Flask API (Render), which loads trained ML models. Highlight 8 REST endpoints and 6 ML classifiers.
3	Live Feature Demonstration (Amarthala Kanthi Sri)	Walk through each page: - Recommend: Enter N=90, P=40, K=40, Temp=20, Humidity=80, pH=7, Rainfall=100 -> predicts Rice at 28% confidence - Suitability: Same inputs -> Poor suitability with detailed assessment - Dashboard: Model comparison table, Chart.js bar chart, 10 chart images, K-Means card
4	Q&A Session (All team members)	Be prepared to answer: - Why Random Forest over other models? - How was the dataset sourced? - How does K-Means differ from the prediction model? - Deployment and scaling considerations